

Total No. of Questions : 8]

SEAT No. :

PD-4085

[Total No. of Pages : 3

[6402]-45

S.E. (Artificial Intelligence and Data Science)

OPERATING SYSTEM

(2019 Pattern) (Semester - III) (217521)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Solve Questions Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume Suitable data if necessary.

Q1) a) Provide solution to producer- Consumer process problem using semaphore. [6]

b) What is deadlock? What are the conditions under which a deadlock situation may arise? [6]

c) What is banker's algorithm? Explain it with suitable example. [6]

OR

Q2) a) What is Process? Differentiate between Process and Thread? [6]

b) Write a short note on Pipe, Semaphore, Signal. [6]

c) With given matrices, determine safe state with the help of banker algorithm. [6]

Claim Matrix C				Allocation Matrix A				Resource Vector R		
	R ₁	R ₂	R ₃		R ₁	R ₂	R ₃	R ₁	R ₂	R ₃
P ₁	3	2	2	P ₁	1	0	0	9	3	6
P ₂	6	1	3	P ₂	6	1	2			
P ₃	3	1	4	P ₃	2	1	1			
P ₄	4	2	2	P ₄	0	0	2			
Available Vector V : R ₁ -0, R ₂ -1, R ₃ -1										

P.T.O.

Q3) a) Write a note on (Any two) : **[6]**

- i) Demand Paging
- ii) Segmentation
- iii) Memory Partitioning

b) Consider the page reference string 2, 3, 2, 1, 5, 2, 4, 5, 3, 2, 5, 2 Calculate page fault for FIFO. If the number of page frames : 3 **[6]**

c) How operating system utilizes swapping techniques in memory management? Explain in detail. **[5]**

OR

Q4) a) Explain following memory allocation strategies with suitable example. **[6]**

- i) Best fit
- ii) First fit
- iii) Next fit

b) What is fragmentation? Explain types of fragmentation with example. **[5]**

c) Given memory partition of 100K, 500K, 200K, 300K and 600K (in order). How would each of First fit, Best fit and worst fit algorithm place processes of size 212K, 417K, 112K, 426K (in order)? Which also makes the most efficient use of memory. **[6]**

Q5) a) List and Explain file types and file access methods. **[6]**

b) List methods of allocation of disk space. Explain any of these methods. **[6]**

c) What information is present in directories. Explain the structure of directories in detail. **[6]**

OR

- Q6)** a) Explain the following concepts [6]
- i) Disk Access methods
 - ii) Directory structure
- b) Explain the advantages of organizing file directory structure into a tree structure? [6]
- c) List and explain various layers of a file system. [6]
- Q7)** a) What is make utility? Explain it with example. Consider your own make file. [6]
- b) Explain share memory with system calls. [5]
- c) Give overview of Linux operating system. Also explain its goals. [6]

OR

- Q8)** a) Define the components of LINUX system with diagram. What is the responsibility of kernel in LINUX operating system? [6]
- b) What are different Process management system calls in Linux. Explain any two with example. [6]
- c) What are the primary goals of conflict-resolution mechanisms used by the Linux kernel for loading kernel modules? [5]
